

Pharmacovigilance — ICSR Case Intake

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 02 of 8 · triage, de-duplicate, MedDRA-code, and draft E2B narratives — a safety physician validates the case

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM CASE BACKLOG TO SAME-DAY TRIAGE

Pharmacovigilance — ICSR Case Intake

A governed ICSR copilot that triages intake, detects duplicates, proposes MedDRA coding, and drafts the E2B(R3) narrative — while a safety physician validates causality and the reportable case before it leaves the building.

~28M

cumulative FAERS reports by end-2023;
quarterly volume ~3× since 2007

[gov/peer-reviewed — FDA FAERS]

40–85%

of PV budgets consumed by case intake &
processing

[sector-press/estimate]

40–70%

manual data-entry time cut in a Pfizer AI/RPA
pilot

[gov/peer-reviewed — Schmider 2019]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Adverse-event volumes scale with portfolio breadth; intake is high-volume and time-critical.
- Triage, duplicate detection, MedDRA coding, and E2B narrative drafting are repetitive and regulated.
- Case processing consumes up to two-thirds of internal PV resources; median ~69 min per report.
- ICSR volumes are growing ~8–15%/year — the backlog compounds without automation.

THE COST OF DOING NOTHING

~\$2.0M / yr

Modeled: 100,000 cases/yr × ~\$20 outsourced intake/case = ~\$2.0M/yr before internal labor. Intake/processing is 40–85% of PV budget; ~8–15% annual volume growth.

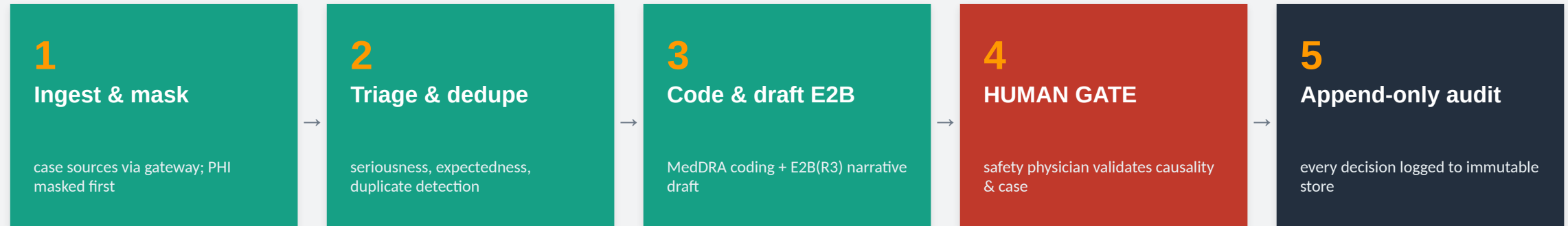
- Late or missed expedited reporting — regulatory and patient-safety exposure.
- Coding inconsistency that undermines signal detection across the portfolio.

[modeled + sector-press/estimate]

The design line: the agent triages, de-duplicates, codes, and drafts; a qualified safety physician validates causality and the reportable case before submission.

How We Solve It — A Governed Pipeline

Ingest case data through a scoped gateway, propose triage / dedupe / MedDRA coding and draft the narrative — then stop at a safety-physician gate before reporting.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

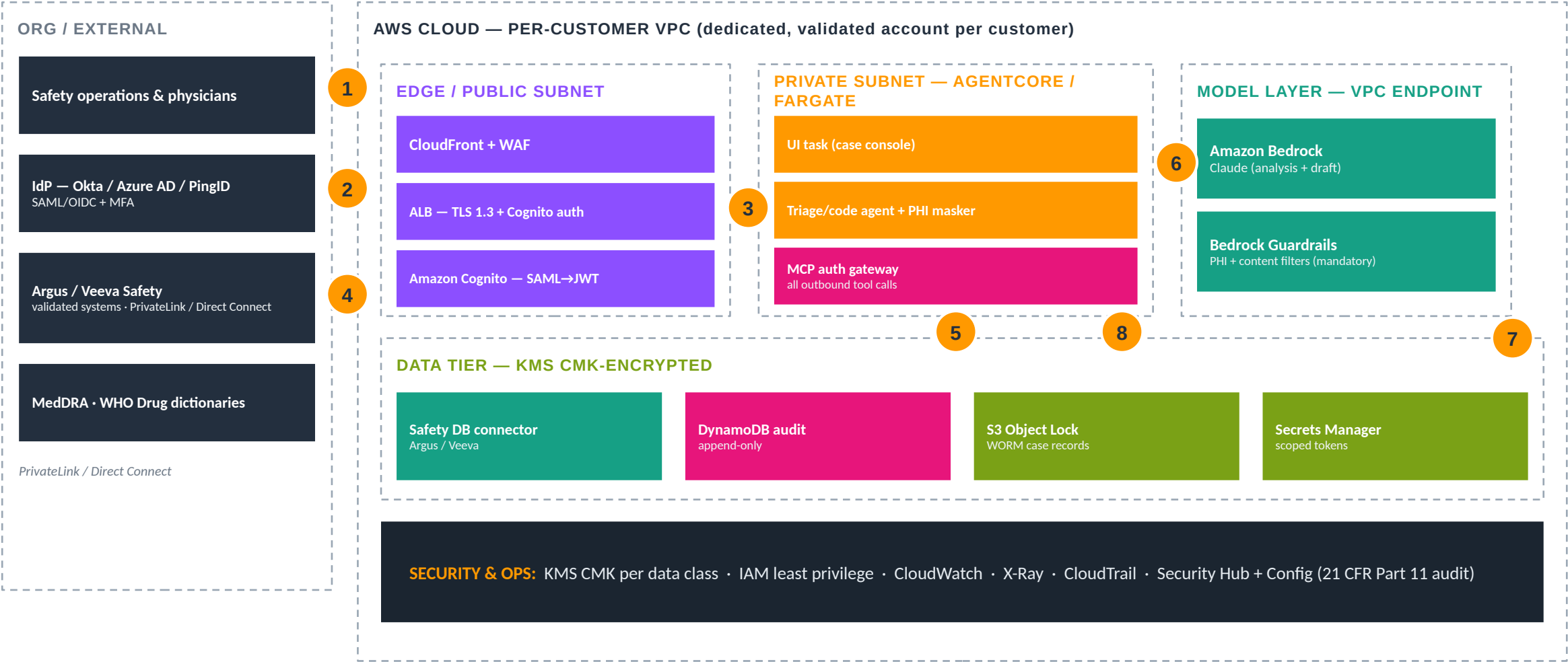
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides the reportable case

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 safety-ops sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to case console 4 safety DB + dictionary reads over private connectivity 5 case volume scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails (PHI masked first) 7 every triage/coding/causality decision persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

40–70%

manual data-entry time cut (Pfizer pilot)

[gov/peer-reviewed — Schmider]

up to 2/3

of internal PV resources today go to case processing

[gov/peer-reviewed]

10–30%

PV cost savings with GenAI/automation deployed

[industry-research]

~69 min

median processing per report today — the target

[gov/peer-reviewed]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (safety-ops / physician roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect Argus/Veeva Safety + MedDRA/WHO Drug; enable PHI masker.
1. Enable S3 Object Lock + append-only audit; run the live end-to-end demo path.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; Agent 02 ships a live Bedrock + real-connector reference path you can run end-to-end.

aws-native-reference/02-pharmacovigilance/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.